Recurrence for $\lambda(n)$ [edit]

The Carmichael lambda function of a prime power can be expressed in terms of the Euler totient. Any number that is not 1 or a prime power can be written uniquely as the product of distinct prime powers, in which case λ of the product is the least common multiple of the λ of the prime power factors. Specifically, $\lambda(n)$ is given by the recurrence

$$\lambda(n)=\begin{cases}\varphi(n) & \text{if } n \text{ is 1, 2, 4, or an odd prime power,}\\ \tfrac{1}{2}\varphi(n) & \text{if } n=2^r,\ r\geq 3,\\ \operatorname{lcm}\Bigl(\lambda(n_1),\lambda(n_2),\ldots,\lambda(n_k)\Bigr) & \text{if } n=n_1n_2\ldots n_k \text{ where } n_1,n_2,\ldots,n_k \text{ are powers of distinct primes.}\end{cases}$$

Euler's totient for a prime power, that is, a number p^r with p prime and $r \geq 1$, is given by

$$\varphi(p^r)=p^{r-1}(p-1).$$